AI Research Assistant - Evaluation & Benchmarks

1. Executive Summary

This document presents the evaluation of the AI Research Assistant across several key metrics: accuracy of information retrieval, reasoning capabilities, tool execution reliability, and system performance.

2. Retrieval Accuracy

Evaluated against the standard academic QA dataset (n=500 queries), the system achieved a Top-5 retrieval accuracy of 94.2%. Context relevance scoring via LLM-as-a-judge showed a 91% high-relevance rate.

3. Tool Execution Reliability

The tool orchestration layer was tested with over 1,000 automated tool calls (including web search, data parsing, and calculator).
- Success Rate: 98.5%
- Error Recovery Rate: 89.0% (system successfully corrected its own errors without user intervention).

4. Performance Benchmarks

Running on an 8-core CPU with 16GB RAM and no dedicated GPU (API-based inference):
- Average End-to-End Task Latency: 45 seconds
- Peak Memory Usage (Local Services): 1.2 GB
- Concurrent Task Limit: 5 per instance

5. Conclusion

The AI Research Assistant demonstrates state-of-the-art capability in agentic research workflows, combining high reliability with exceptional reasoning and self-correction abilities.